



Indian Institute of Information Technology, Surat
Department of Computer Science & Engineering
Mid Semester Examination, 4th Sem, 2024-25
Design Analysis and Algorithm (CS 405)

Timing: 1:30 PM to 3:00 PM

Date: 07/03/2025

Max Marks: 30

- Note: 1) Attempt all parts of a question at one place.
2) Make necessary assumptions wherever required.

1. What do you mean by Asymptotic Notations? What are their needs? Explain the Asymptotic notations with diagram for the following: (6)

- i. Big-Oh ii. Big-Omega iii. Theta Notation

2. Consider three algorithms A, B and C for solving the same problem. (5)

(a) Algorithm A solves the problem by dividing it into eight subproblems of half the size, and recursively solving each subproblem, and then combining the solutions in linear time. Write a recurrence relation for the running time $T(n)$ of algorithm A for an input of size n .

(b) Solve the recurrence relation obtained above using the Master method. State which case of the Master method was used.

(c) Algorithm B solves a problem of size n by recursively solving two subproblems of size $n - 1$ and then combining their solutions in constant time. Write a recurrence relation for the running time $T(n)$ of algorithm B for an input of size n . Express the running time of algorithm B using big-O notation.

(d) Algorithm C solves a problem of size n by dividing it into 9 subproblems of size $n/3$, recursively solving each subproblem, and then combining the solutions in $O(n)$ time. What is the running time of Algorithm C?

(e) Which of the three algorithms would you choose for this problem?

3. Write pseudocode with an example for the following sorting algorithms: i) Quick Sort and ii) Merge Sort. Argue on its best case, average case and worst case time complexity. (7)

4. Coins in the United States are minted with denominations of 1, 5, 10, 25 and 50 cents. Now consider a country whose coins are minted with denominations of $d_1, \dots, d_k$ units. We seek an algorithm to make change of n units using the minimum number of coins for this country. For example, if the set of denominations is $[1, 5, 10]$, then the minimum number of coins needed to make change for 26 is 4 because $26 = 10 + 10 + 5 + 1$ and 26 cannot be expressed as a sum of fewer than 4 coins. We want to design an algorithm $\text{mincoins}(d, n)$ which takes as input a set of denominations $d = [d_1, \dots, d_k]$ and a change amount n , and which outputs the minimum number of coins needed to obtain the change amount n in terms of denominations in d . (2)

(a) The greedy algorithm repeatedly selects the biggest coin no bigger than the amount to be changed and repeats until it is zero. Show that the greedy algorithm does not always use the minimum number of coins in a country whose denominations are $[1, 6, 10]$.

5.

Suppose there are 10 activities, each with a specified start and end time, as given below:

(5)

$S = (A1 A2 A3 A4 A5 A6 A7 A8 A9 A10)$

$S_i = (1, 2, 3, 4, 7, 8, 9, 9, 11, 12)$

$F_i = (3, 5, 4, 7, 10, 9, 11, 13, 12, 14)$

Determine a schedule that maximizes the number of activities, explaining each step in the associated algorithm.

6.

Given the jobs, their deadlines and associated profits as shown-

(5)

Jobs	J1	J2	J3	J4	J5	J6
Deadlines	5	3	3	2	4	2
Profits	200	180	190	300	120	100

Implement the Job Sequencing Algorithm and:

(i) Identify the optimal job schedule to achieve maximum profit.

(ii) Compute the total maximum profit earned.

****End****